X-ray applications and CT

When does a projection answer the question?

A projection is in single view of x-ray absorption.

- **Chest radiography:** one projection can show the overall lung pattern and the position of tubes and lines.
- **Mammography:** a projection can show fine calcifications when there is enough contrast and resolution.
- **Fluoroscopy:** repeated projections show motion during a procedure.

When the question is **where within the body** a feature lies, overlapping structures can make a single view ambiguous.

A projection versus a slice

Chest radiograph

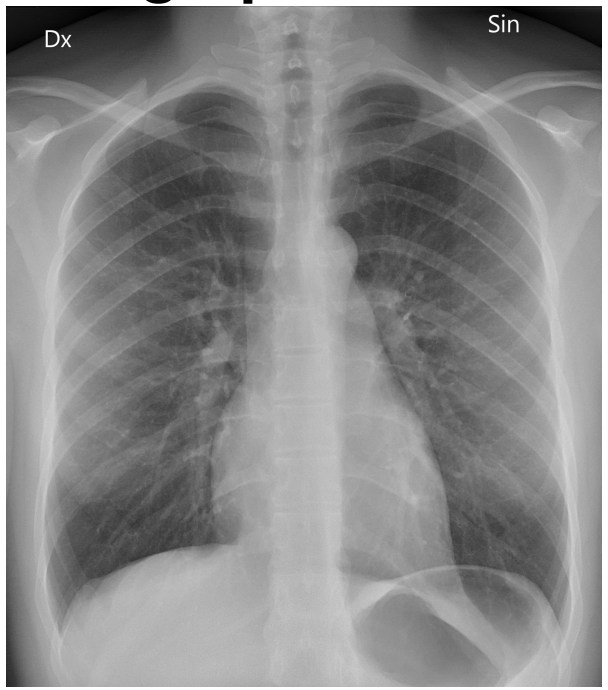


Figure 1: Normal chest radiograph

Chest CT slice

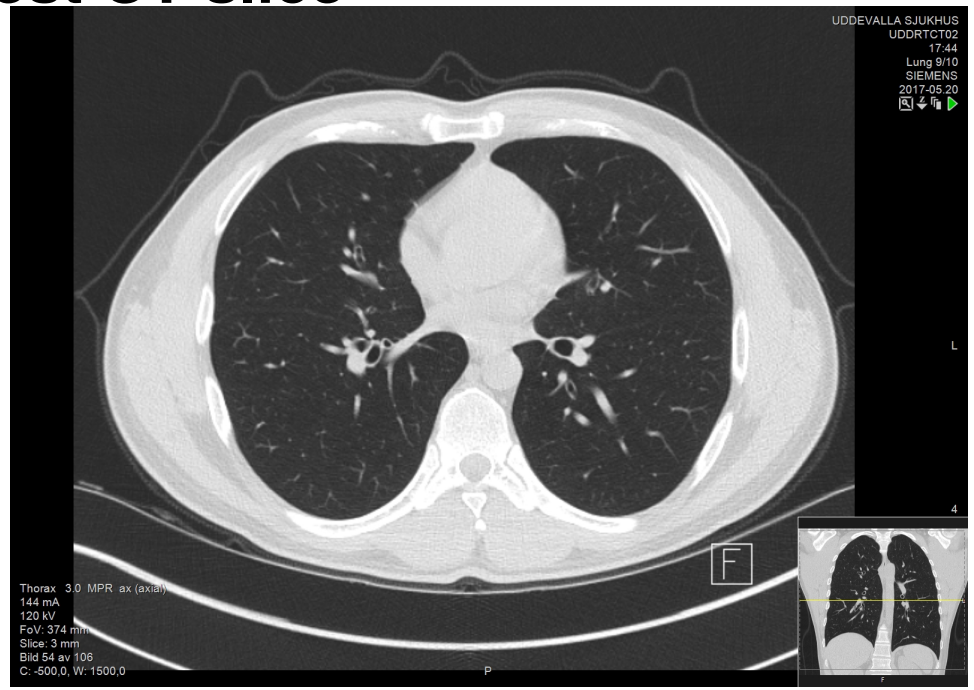


Figure 2: Axial CT slice of the chest

The projection combines structures along each ray. The slice shows their positions within one cross-section.

A CT scanner acquires views around the body

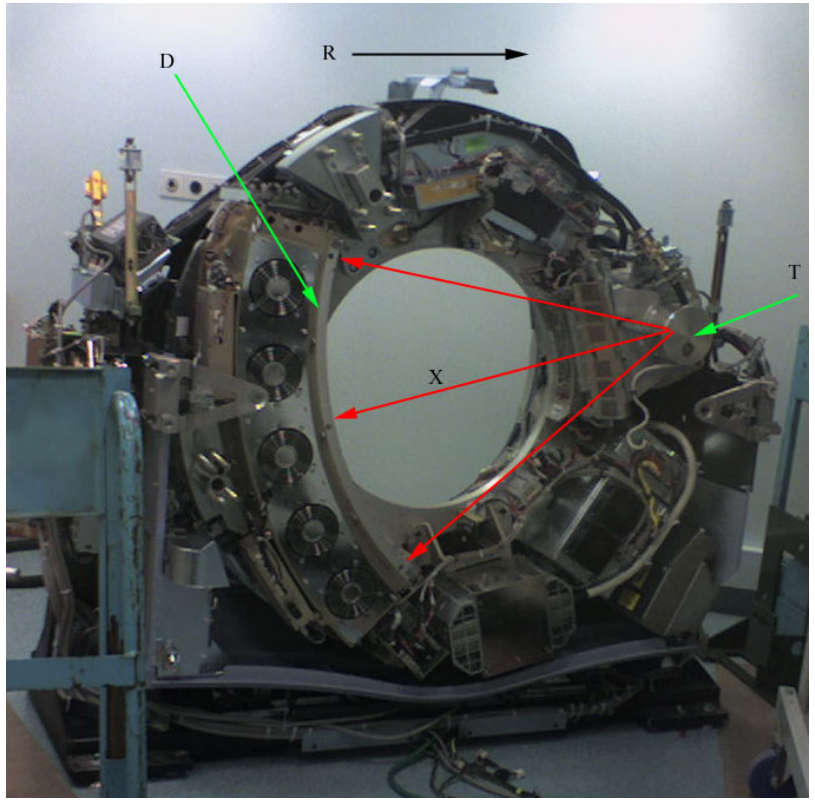


Figure 3: CT gantry with housing removed and components labeled

The **gantry** supports the X-ray tube and detector array. In this scanner they rotate together around the patient.

- **T:** tube
- **D:** detector array
- **X:** fan-shaped beam
- **R:** direction of rotation

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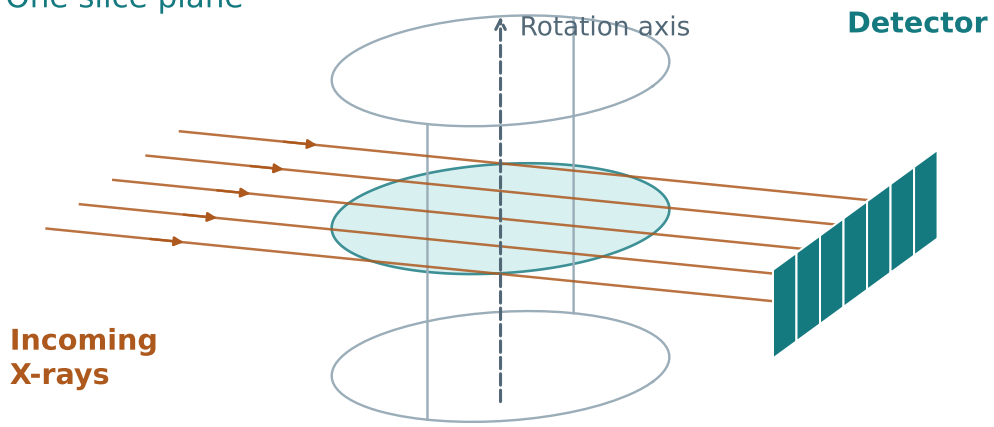
One detector row samples a slice

How detector rows sample a CT slice

One view of an ideal parallel-beam scan

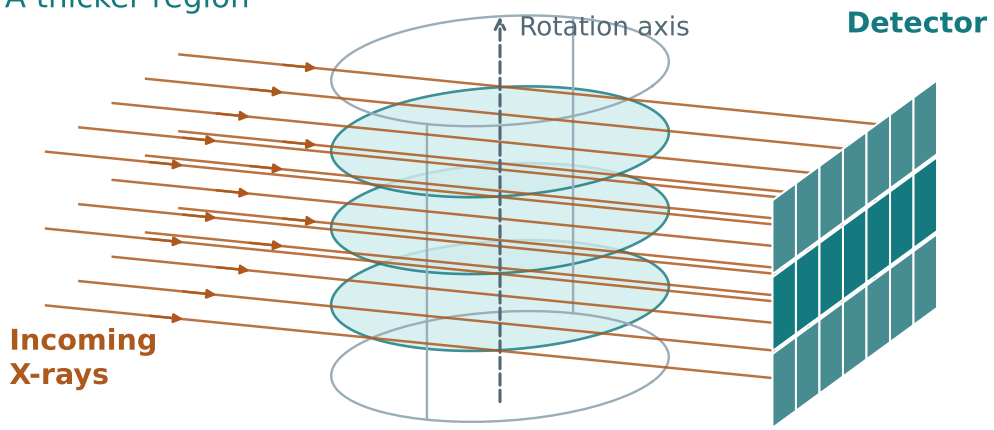
One detector row

One slice plane



Multiple detector rows

A thicker region



The rays travel within the slice plane. As the view rotates, the rays change direction within that same plane.

The right panel extends the parallel-beam model. Clinical multirow CT uses a beam that diverges across detector rows.

Each ray measures a sum of attenuation along its path

A **ray** is unique direction of x-ray travel.

Recall the two-layer result:

$$-\ln(I/I_0) = \mu_1 x_1 + \mu_2 x_2.$$

For a ray through a slice with continuously varying attenuation,

$$p(s,\theta) = -\ln(I/I_0) = \int_{\text{ray}} \mu(x,y)\, d\ell.$$

This is a **line integral**: add the local attenuation coefficient times a short distance along the ray.

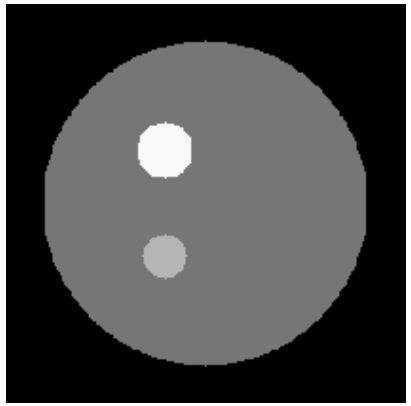
Ray position on the detector

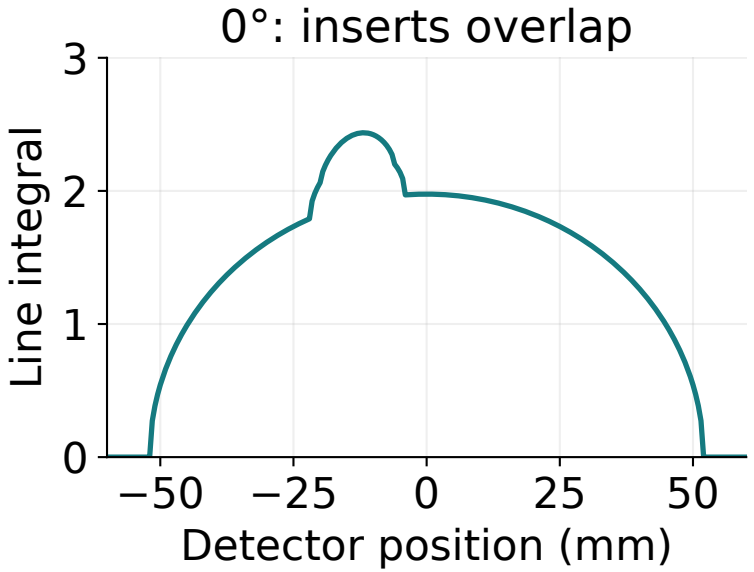
- s is the signed position along the detector row, in mm. The central ray has $s = 0$.
- θ is the angle of the detector coordinate measured from the x axis. The rays run perpendicular to that coordinate.
- The ray through (x, y) reaches detector position $s = x \cos\theta + y \sin\theta$.

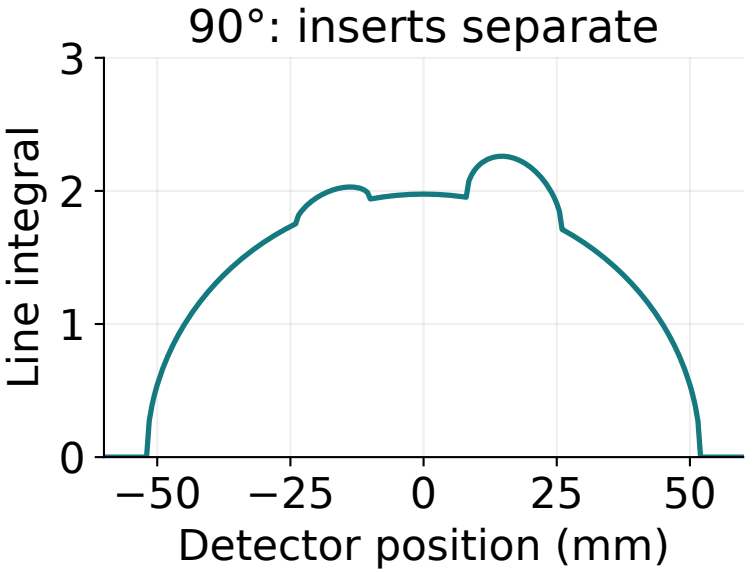
At $0°$, $s = x$ and the rays run parallel to y. At $90°$, $s = y$ and the rays run parallel to x.

One view produces a profile across the detector

Slice with two inserts

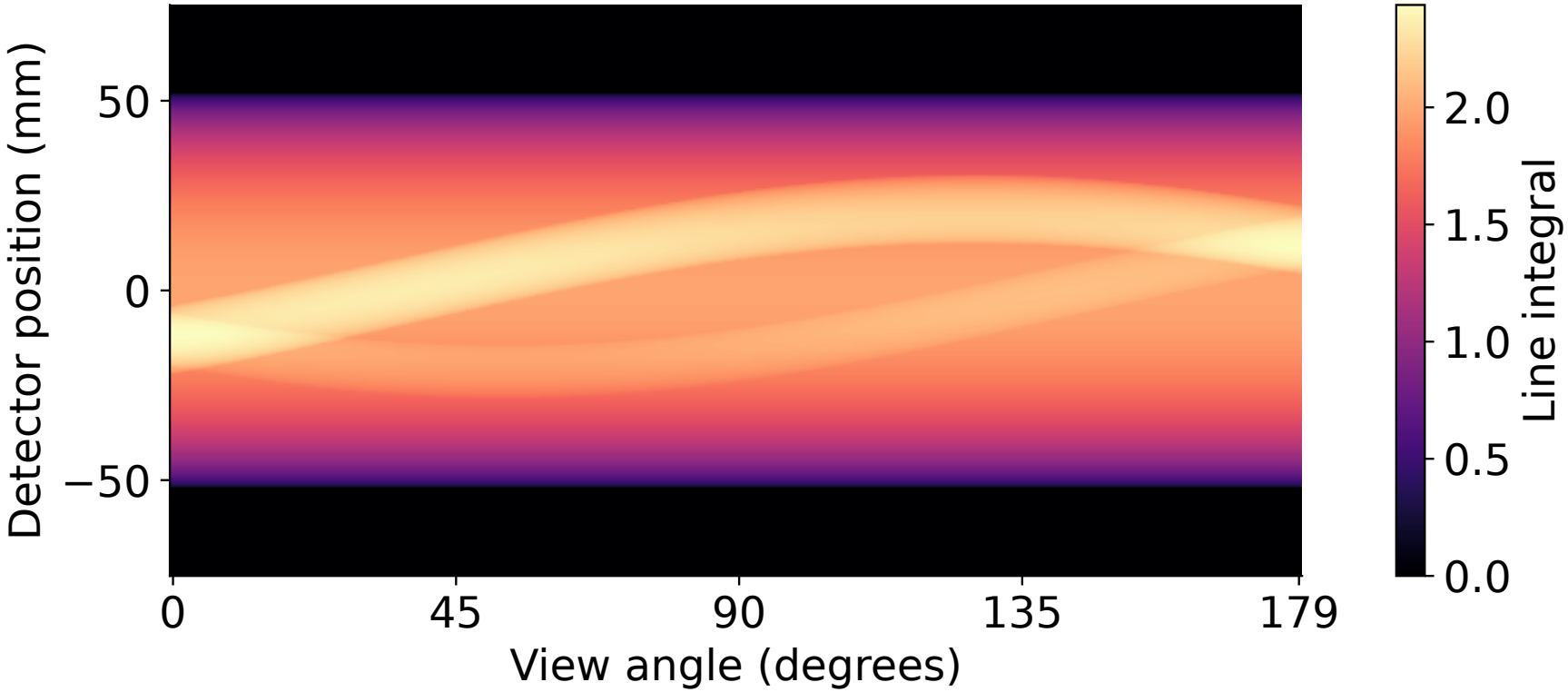






Each point is a line integral of the linear absorption coefficient for a different ray through the same slice. One complete detector row gives one profile.

A sinogram stacks profiles by angle



Each column is one detector profile $p(s, \theta)$. Its values are dimensionless.

Follow the inserts. How does the position on the detector change as the view rotates?

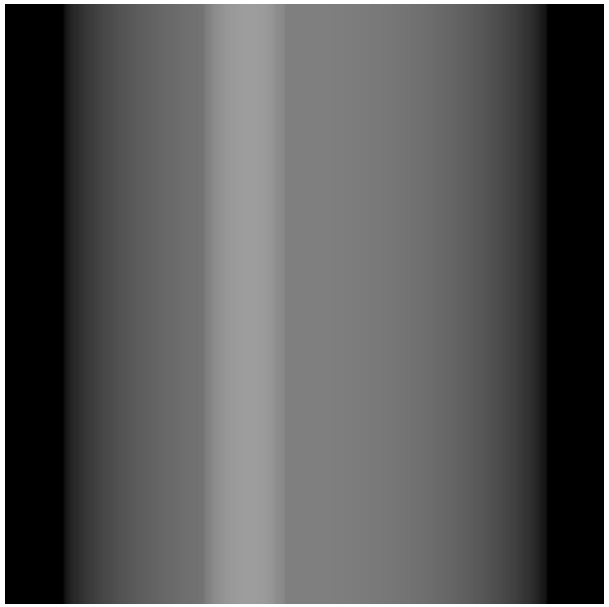
Backprojection spreads each value along its ray

For each detector measurement, add its value back along the ray that produced it.

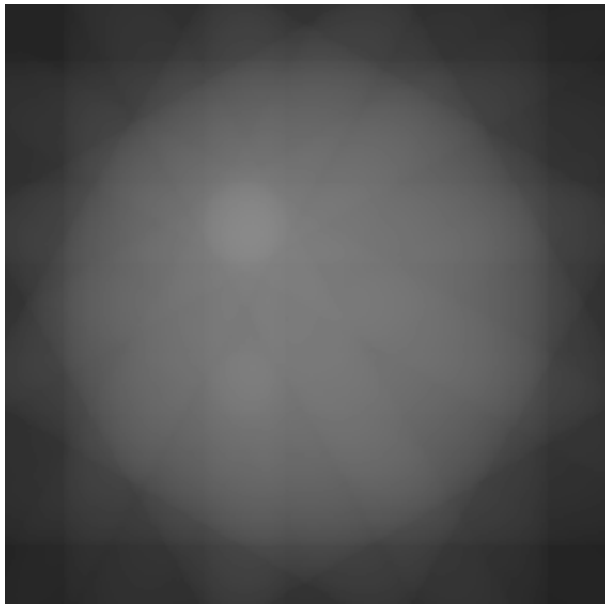
A single view gives stripes of possible locations. As views accumulate, rays from the same structure cross near that structure.

More views concentrate the signal

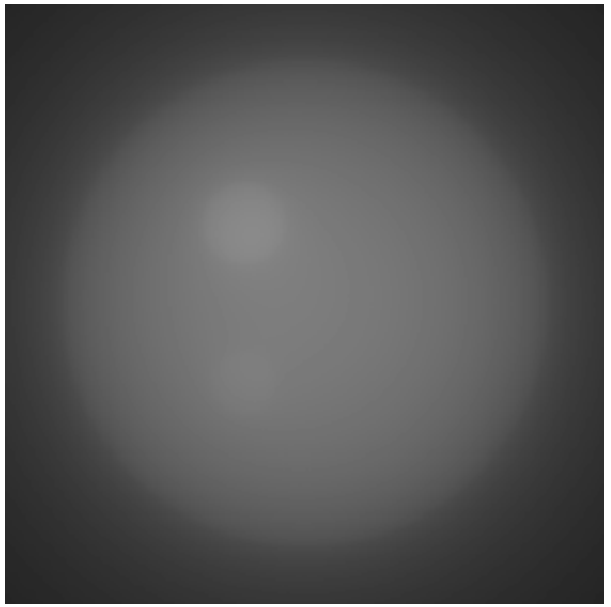
1 view



6 views



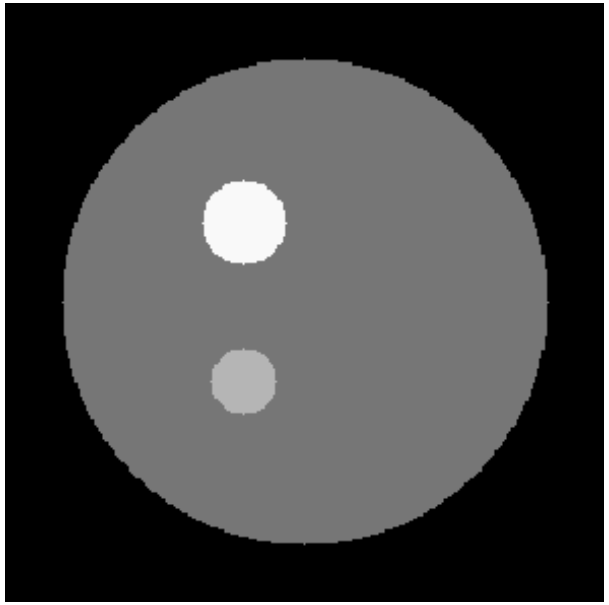
60 views



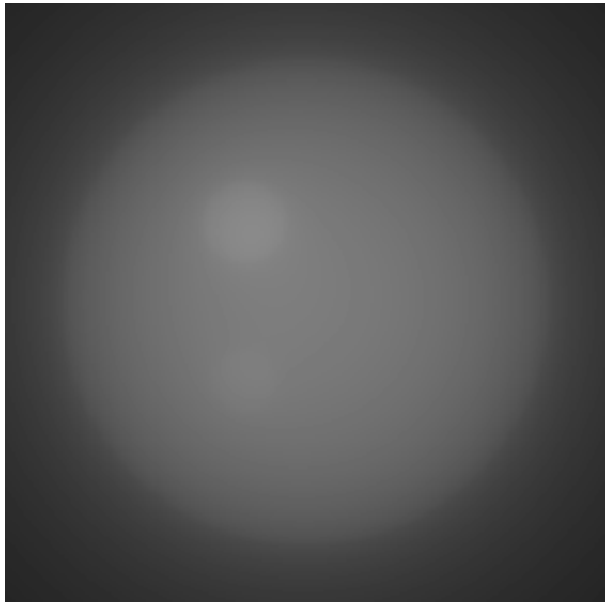
The inserts become easier to locate as views accumulate. Their edges still look broad.

Filtering sharpens the reconstructed features

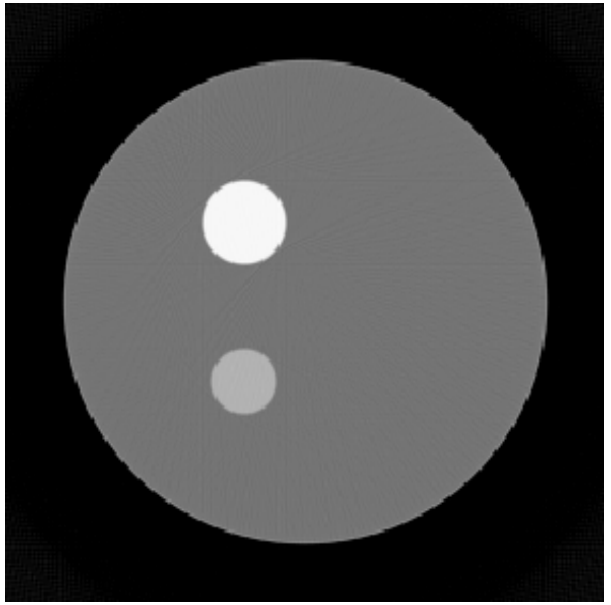
Known object



Unfiltered backprojection



Filtered backprojection



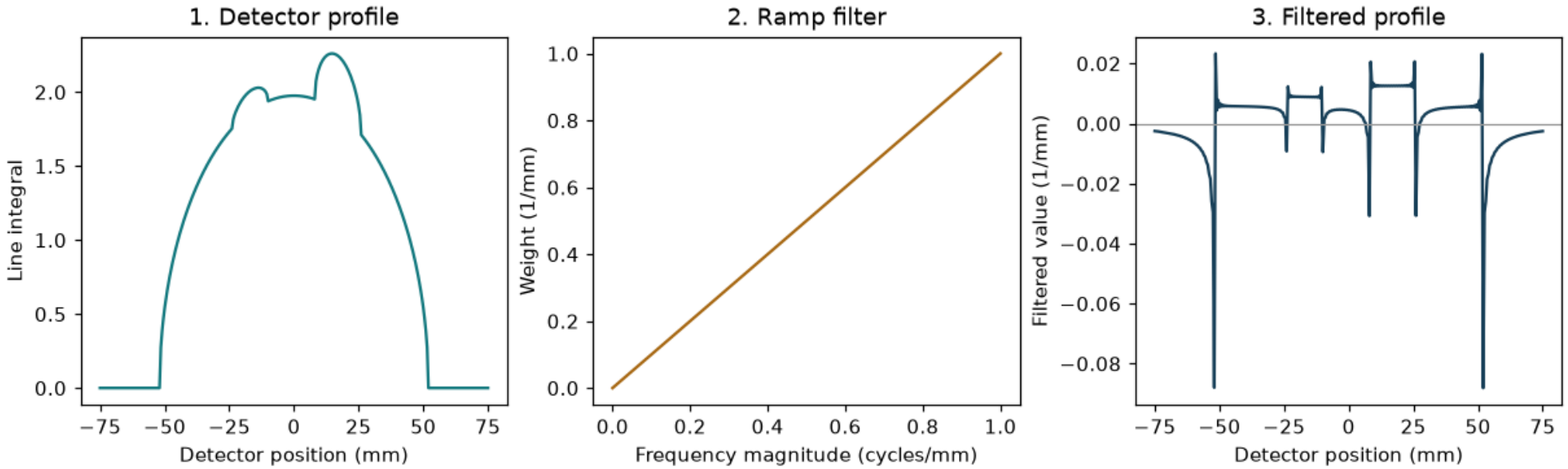
Filtered backprojection filters each detector profile before spreading its values back along the rays.

Unfiltered brightness is scaled to match the reference background.

Filter across detector position

For a fixed view θ, Fourier transform the detector profile **along** s. The spatial frequency u is in cycles/mm.

$$P(u,\theta) = \mathcal{F}_s\{p(s,\theta)\}, \qquad q(s,\theta) = \mathcal{F}_s^{-1}\{|u|P(u,\theta)\}.$$

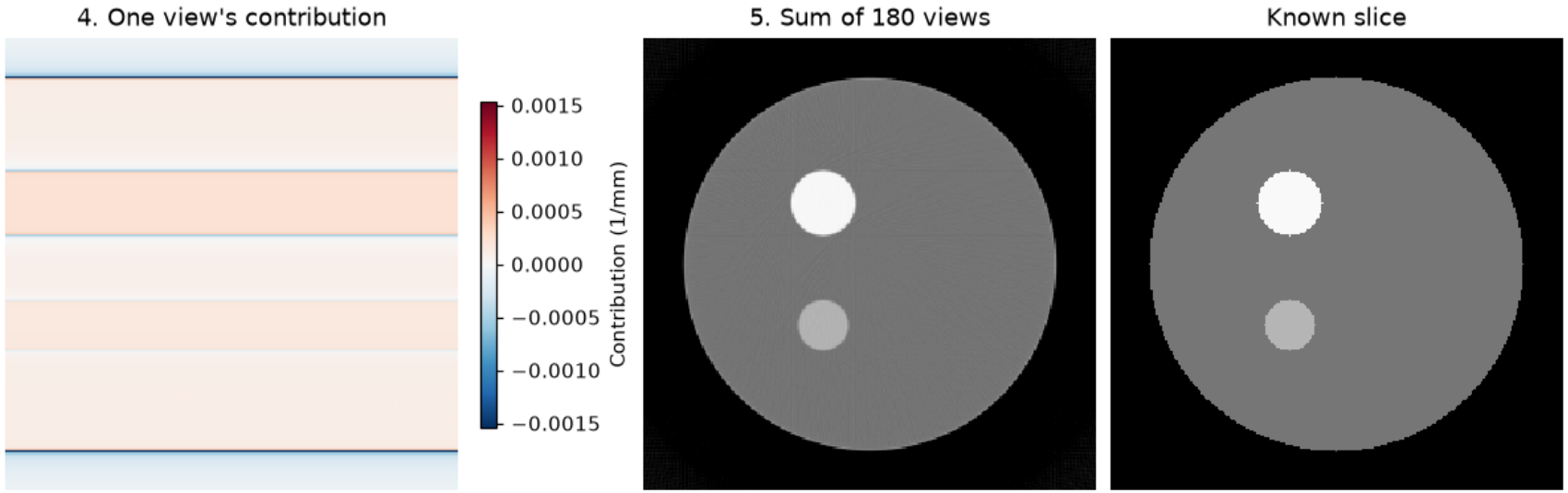


The ramp $|u|$ gives higher spatial frequencies more weight. Apply it to each sinogram column independently.

What does one view add to the slice?

For each pixel (x, y), sum the filtered values along the rays through it.

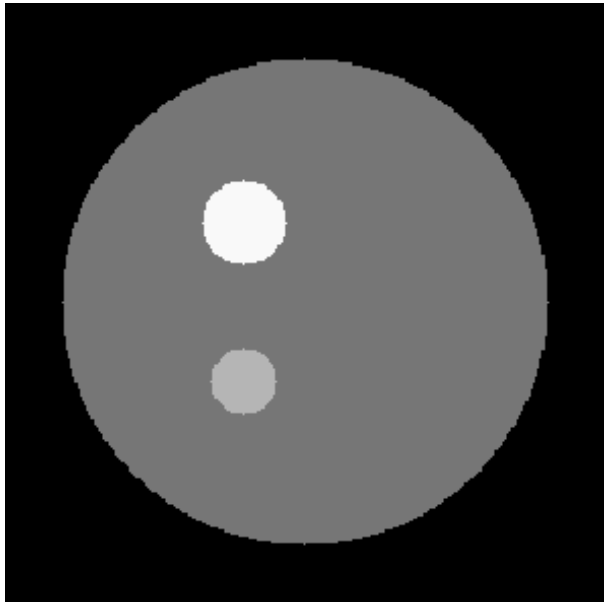
$$\hat{\mu}(x, y) \approx \Delta\theta \sum_{j=1}^{N} q(x \cos\theta_j + y \sin\theta_j, \theta_j), \qquad \Delta\theta = \pi / N.$$



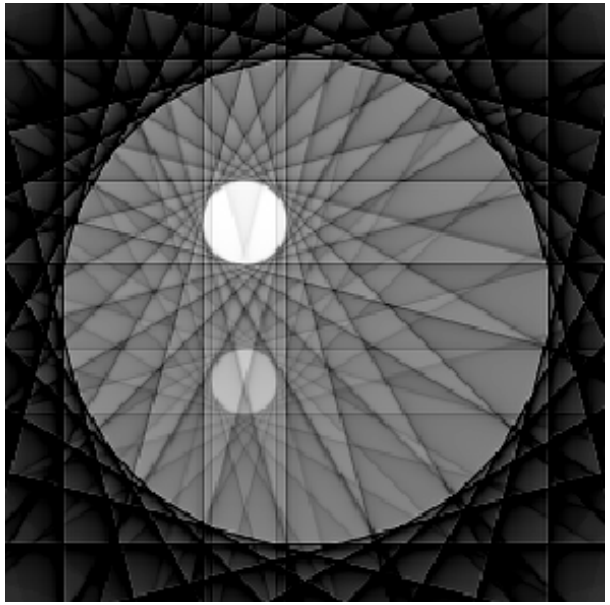
At $90°$, $s = y$ and the rays run horizontally. The left panel shows signed contributions in mm^{-1}. Each panel spans -63 to $+63$ mm along x and y.

Too few views leave streaks

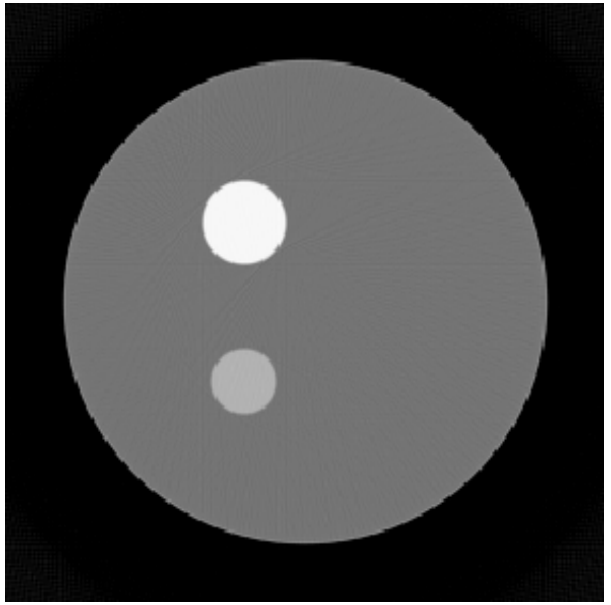
Known object



12 views



180 views



Both images use filtered backprojection.

What does a CT pixel represent?

A reconstructed CT slice estimates the linear attenuation coefficient at locations within the slice.

Clinical images commonly express attenuation relative to water using **Hounsfield units (HU)**. The display maps a selected range of values to grayscale.

What is the difference between a detector reading in one view and a pixel in a reconstructed slice?

Reading and image sources

IAEA, *Diagnostic Radiology Physics*, chapters 6 and 11.

The projection and reconstruction examples use a synthetic two-insert object and an ideal parallel-beam model.

CT gantry photograph by ChumpusRex · CC BY-SA 3.0.